NARAYANA
IIT - JEE ACADEMY - INDIA

SEC:: SR. JEE MAINS APEX | GTM-18(FN) | Date: 16.01.2025
Time: 3.00Hrs. | JEE MAINS -2025 | Max. Marks: 300 M

KEYSHEET

PHYSICS

1	2	3	4	5	6	7	8	9	10
A	C	B	A	D	A	C	D	B	B
11	12	13	14	15	16	17	18	19	20
A	C	B	B	B	B	A	C	C	A
21	22	23	24	25					
2	20	7	8	9					

CHEMISTRY

26	27	28	29	30	31	32	33	34	35
A	B	D	A	A	D	A	D	D	A
36	37	38	39	40	41	42	43	44	45
C	A	D	A	D	B	D	D	D	B
46	47	48	49	50					
5	5	10	5	3					

MATHEMATICS

51	52	53	54	55	56	57	58	59	60
C	B	A	D	C	A	A	C	A	B
61	62	63	64	65	66	67	68	69	70
B	D	D	B	B	C	A	A	B	A
71	72	73	74	75					
5	12	102	960	2					

For More Material Join: @JEEAdvanced_2025

PHYSICS

1. 29 MSD =30 VSD
Smallest MSD = 0.5°
LC=1 MSD = 1VSD

$$\text{LC} = 1\text{MSD} - \frac{29}{30}\text{MSD}$$

$$= \frac{1}{30}\text{MSD} = \frac{1}{30}\times\frac{1}{2} = \left[\frac{1}{60}\right]^{\circ} = 1\text{minute}$$

2. $W = \mu mg\cos\theta S = 0.5\times1\times9.8\times\frac{1}{2}\times1 = 2.45\ J$

3. Stopping potential does not depend upon intensity of incident light (I).

4. Mass of ice melted = 600 – 550 = 50g

$$M'\times0.1\times350 = 600\times0.5\times10 + 50\times80$$

$$M' = 200g$$

$$M' = M\times100g$$

$$M = 2$$

5. $$\frac{3}{2}KT = 7.7\times10^{-14}$$

$$\Rightarrow T = \frac{7.7\times10^{-14}\times2}{3\times1.38\times10^{-23}}$$

$$T = 3.7\times10^{9}\,K$$

6. In an adiabatic process

$$pV^{\gamma} = \text{constant}$$

$$\Rightarrow \frac{p_1}{p_2} = \left(\frac{V_2}{V_1}\right)^{\gamma}$$

$$\Rightarrow \frac{p_1}{p_2} = \left(\frac{1}{8}\right)^{5/3} \quad \Rightarrow \frac{p_1}{p_2} = \left(\frac{1}{2^3}\right)^{5/3} = \frac{1}{32}$$

$$\therefore \frac{p_2}{p_1} = 32$$

7. By using $E = -\frac{13.6}{n^2}eV$ (for H_2 atom)

$$\Rightarrow -0.544 = -\frac{13.6}{n^2} \Rightarrow n^2 = 25 \Rightarrow n = 5$$

$\therefore$ Angular momentum = $n\frac{h}{2\pi} = \frac{5h}{2\pi}$.

For More Material Join: @JEEAdvanced_2025

8. $R = 10\Omega$,

$X_L = \omega L = 2000 \times 5 \times 10^{-3} = 10\Omega$

$X_C = \dfrac{1}{\omega C} = \dfrac{1}{2000 \times 50 \times 10^{-6}} = 10\Omega$

$Z = \sqrt{R^2 + \left(X_L - X_C\right)^2}$

Thus, $Z = 10\Omega$

$i_0 = \dfrac{V_0}{Z} = \dfrac{20}{10} = 2A$; $i_{rms} = \dfrac{2}{\sqrt{2}} = 1.4\ A$

$V_{rms} = 4 \times i_{rms} = 5.6\ A.V.$

9. $y = \dfrac{x^3}{6}$

$\dfrac{dy}{dx} = slope = \tan\theta = \mu = \dfrac{3x^2}{6}$

$x = 1m$

$y = \dfrac{1}{6}m$

11. n is number of polarides then $I_f = \dfrac{I_0}{2}\left(\cos^2\theta\right)^{n-1}$

12. $\dfrac{-Gmm'}{r} \times 3 + \dfrac{1}{2}m'v_e^2 = 0$

Or $\dfrac{-3Gm}{\left(\dfrac{a}{2} / \cos 30^0\right)} + \dfrac{1}{2}v_e^2 = 0$

$v_e = \sqrt{\dfrac{6\sqrt{3}Gm}{a}}$

13. $\dfrac{1}{4\pi \in_0}\dfrac{Qq}{r^2} = \dfrac{mv^2}{r}$,

Or $v = \sqrt{\dfrac{Qq}{4\pi \in_0 mr}}$

Now time period, $T = \dfrac{2\pi r}{v} = \dfrac{2\pi r}{\sqrt{\dfrac{Qq}{4\pi \in_0 mr}}}$

Or $r^3 = \left(\dfrac{Qq}{16\pi^3 \in_0 m}\right)T^2$

For More Material Join: @JEEAdvanced_2025

14. We know that,

$$F = \frac{1}{4\pi \in_0}\frac{q_1 q_2}{r^2}$$

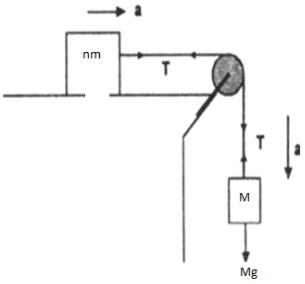
$$\therefore \in_0 = \frac{1}{4\pi \in_0}\frac{q_1 q_2}{r^2}$$

$$= \frac{1}{MLT^{-2}} \times \frac{(AT)^2}{L^2}$$

$$= M^{-1}L^{-3}T^4A^2$$

16. $$\gamma = \frac{n_1 Cp_1 + n_2 Cp_2}{n_1 Cv_1 + n_2 Cv_2}$$

17.



$$Mg - T = Ma$$

For 'n' blocks T = nma

$$a = \frac{Mg}{nm + M}$$

$$T_n = ma = \frac{mMg}{nm + M}$$

18. When bulbs are connected in series, $P = \frac{V^2}{R'} = \frac{V^2}{3R}$

When bulbs are connected in parallel,

$$P' = \frac{V^2}{R''} = \frac{V^2 \times 3}{R} = 3 \times 3P = 9P.$$

19. Conceptual

20. $$v_0 \cos\theta = \frac{v_0}{2}$$

$$\cos\theta = \frac{1}{2} \Rightarrow \theta = 60^0$$

21. $$I_P N_P = I_S N_S$$

For More Material Join: @JEEAdvanced_2025

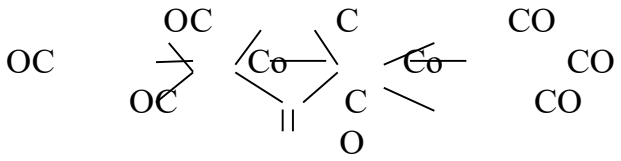
22. $f = \frac{R}{2\mu} = \frac{30}{2\times1.5} = 10cm$
Object should be placed at (2f) of the concave mirror
[to get image of object size]
So, distance of object from lens is $2f = 20cm$

23. $B = \frac{\mu_o I}{4\pi d}[\sin\theta_1 + \sin\theta_2]$

24. $E \propto \frac{1}{r^3},\ F \propto E \Rightarrow F \propto \frac{1}{r^3}$

25. $S = \frac{I_g(G)}{I - I_g}$

CHEMISTRY

26. 

27. $I_3^- - sp^3d$, *Linear*
$XeF_2 - sp^3d$, *Linear*

28. Conceptual

29. Conceptual

30. SRP values of $\underset{1.74V}{Bro4^-}$ $\underset{1.19V}{Clo4^-}$ $\underset{1.65V}{Io4^-}$

31. Conceptual

32. $number\ of\ moles\ of\ He = 4/4 = 1$ $number\ of\ moles\ of\ CH_4 = \frac{5.6}{22.4} = 0.25$

$number\ of\ moles\ of\ CO_2 = \frac{3\times10^{22}}{6\times10^{23}} = 0.05$

$mole\ fraction\ of\ He = \frac{1}{1.30}$

33. Conceptual

34. B.P of n-pentane > 2-methyl butane > 2, 2 -dimethyl propane >n-butane

35. Cro_5 has two peroxy bonds
$x + 4(-1) + 1(-2) = 0 \quad \Rightarrow x = 6$

36. It depends on number of particles

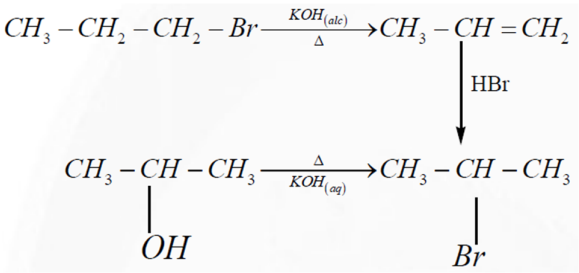
37. Conceptual

39. Conceptual

40. One with low SRP reduces the metal / Non metal with high SRP.

43.

For More Material Join: @JEEAdvanced_2025



46. Five

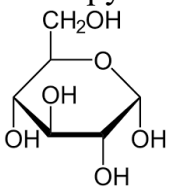
47. Be, Mg, Co, Ni – do not gives, colour in flame test

48.

$$Mg(OH)_2 \rightleftharpoons Mg^{2+} + 2OH^- \qquad K_{sp} = \left[Mg^{2+}\right]\left[OH^-\right]^2 \qquad \left[OH^-\right] = \sqrt{\frac{K_{sp}}{\left[Mg^{2+}\right]}} = 10^{-4}$$

$pOH = 4$ and $pH = 10$

49. Glucopyranose has 5 chiral centres



50. 3 moles of Acetone gives phorone

MATHEMATICS

51. $\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 2 & 5 & \lambda \end{vmatrix} \neq 0$

52. $\left(m^2 n\right)^{1/3} = \cos\theta, \sin\theta = \left(mn^2\right)^{1/3}$

53.

$\bar{a} \times \left(\bar{b} - 3\bar{c}\right) = \bar{0}$

$\left|\bar{b} - 3\bar{c}\right| = \lambda\left|\bar{a}\right|$

$\Rightarrow \lambda^2 = 1$

54. $b = \frac{1}{3}(2ae) \Rightarrow \frac{b}{a} = \frac{2}{3}e \Rightarrow e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{4e^2}{9}}$

$\Rightarrow 9e^2 = 9 - 4e^2 \Rightarrow 13e^2 = 9 \Rightarrow e = \frac{3}{\sqrt{13}}$

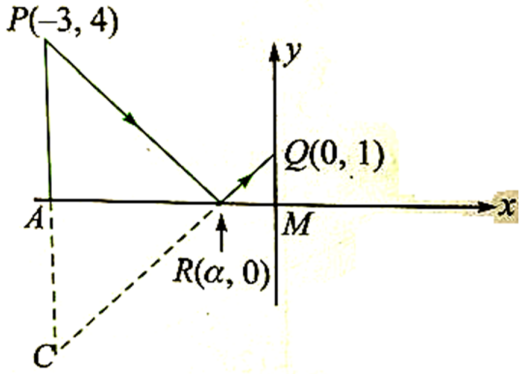
55 $g(x) = 3x^4 + 8x^3 - 18x^2 + 60$

$g'(x) = 0 \quad in[-2, 9]$

For More Material Join: @JEEAdvanced_2025

$\Rightarrow x = 1, 0$

56. For PR = RQ to be minimum, it should be the path of light.



57. $T_3 = LR_1^2 = M \quad , \quad T_{11} = LR_1^{10}$

$LR_2^4 = M$

$LR_1^{10} = LR_2^{k-1}$

$\Rightarrow K = 21$

58. $\left|\bar{b} \times \bar{a} - \bar{b}\right|^2 = \left|\bar{b} \times \bar{a}\right|^2 + \left|\bar{b}\right|^2 = 10 = K$

59. $f'(3) = 1, \quad f'(2) = \dfrac{1}{\sqrt{3}}$

$\alpha = 12, \quad \beta = -10$

60. $S = (-5, 0), S' = (5, 0)$

$\Rightarrow K = 25$

61. $\dfrac{\frac{1}{2} \cdot \frac{9}{13}}{\frac{1}{2}\frac{9}{13} + \frac{1}{2}\frac{10}{13}} = K$

62. Put $e^{-x} = y \therefore -e^{-x}dx = dy$ or $dx = \dfrac{dy}{-y}$ and limits adjusted as 1 to 0.

$\therefore \; I = \int_1^0 [2y]\frac{dy}{-y} = \int_0^1 \frac{1}{y}[2y]dy$

As y varies from 0 to 1, $2y$ varies from 0 to 2

$\therefore \quad [2y] = 0$ or 1

$\therefore \quad I = \int_0^{1/2} \frac{1}{y}[2y]dy + \int_{1/2}^1 \frac{1}{y}[2y]dy$

$= \int_0^{1/2} \frac{1}{y} \cdot 0 dy + \int_{1/2}^1 \frac{1}{y} \cdot 1 dy$ by (A)

$= [\log y]_{1/2}^1 = 0 - \log\frac{1}{2} = \log 2$

For More Material Join: @JEEAdvanced_2025

63. Let $\overline{OA}=\vec{a},\overline{OB}=\vec{b},\overline{OC}=\vec{c}$, then $\vec{a}.\vec{a}\left(\vec{b}-\vec{c}\right).\left(\vec{b}-\vec{c}\right)=\vec{b}.\vec{b}+\left(\vec{c}-\vec{a}\right).\left(\vec{c}-\vec{a}\right)$

$\overline{BA}.\overline{OC}=0$ Hence, $AB\perp OC$ Similarly, $BC\perp OA$ and CA $\perp OB$

64. $f'\left(1^-\right)=f'\left(1^+\right)$ *and* $f\left(1^-\right)=f\left(1^+\right)$

65. $\int\left(x^{11}+x^7+x^3\right)\left(2x^{12}+3x^8+6x^4\right)^{1/4}dx \;=\frac{x^5}{30}\left(2x^8+3x^4+6\right)^{5/4}+c$

66. $\sum_{r=1}^{10}(r-\omega)\left(r-\omega^2\right)=\sum_{r=1}^{10}\left\{r^2-r(-1)+1\right\}$

$=\sum_{r=1}^{10}\left(r^2+r+1\right)=\Sigma r^2+\Sigma r+\Sigma 1$

$=\frac{r(r+1)(2r+1)}{6}+\frac{r(r+1)}{2}+10$

Put $r=10=\frac{10\cdot 11.21}{6}+\frac{10.11}{2}+10=450$

$\therefore E=\sin\left(450\cdot\frac{\pi}{900}\right)=\sin\frac{\pi}{2}=1.$

67. $T_3=60,\ T_4=160,\ T_5=240$

$\Rightarrow n=6,\ x=1,\ a=2$

68. $\left[2\sin x\right]=-2,\left[\cos x\right]=-1$

$\Rightarrow\frac{7\pi}{6}<x<\frac{3\pi}{2}$

$\sin x+\sqrt{3}\cos x=2\sin\left(\frac{\pi}{3}+x\right)$

69. $I=\int_{-\pi/2}^{\pi/2}\log\frac{px^2+qx+r}{px^2-qx+r}+\int_{-\pi/2}^{\pi/2}\log(a+b)dx+\int_{-\pi/2}^{\pi/2}\log|\sin x|\,dx=I_1+I_2+I_3$

Clearly $I_1=0$ as $-\log t=\log\frac{1}{t}$

$\therefore\quad f(-x)=-f(x)$*i.e.* odd function

$I_2=\log(a+b)x=\log(a+b)(\pi/2+\pi/2)$

$=\pi\log(a+b)$

$I_3=2\int_0^{\pi/2}\log|\sin x|\,dx=2\int_0^{\pi/2}\log\sin x dx$

$=2\cdot\frac{\pi}{2}\log\frac{1}{2}=-\pi\log 2$

For More Material Join: @JEEAdvanced_2025

$\therefore \quad I = 0 + \pi \log(a+b) - \pi \log 2 = \pi \log\left(\frac{a+b}{2}\right)$

70. $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} = 0$

71. The curve is $y = x^3 - 2x^2 + x + 5$

72. Required area $= \int_0^2 \left[2^x - \left(2x - x^2\right)\right] dx$

73. ${}^{101}C_1 + {}^{101}C_2 . S_1 + ... + {}^{101}C_{101} . S_{100} = \alpha^k T_{100}$,

$$\Rightarrow {}^{101}C_1 + {}^{101}C_2(1+q) + {}^{101}C_2\left(1+q+q^2\right) + ...{}^{101}C_{101}\left(1+q+...+q^{100}\right) = 2\alpha^k \frac{\left(1-\left(\frac{1+q}{2}\right)^{101}\right)}{(1-q)}$$

$$\Rightarrow {}^{101}C_1(1-q) + {}^{101}C_2\left(1-q^2\right) + ... + {}^{101}C_{101}\left(1-q^{101}\right) = 2\alpha^k \frac{\left(1-\left(\frac{1+q}{2}\right)^{101}\right)}{(1-q)}$$

$$\Rightarrow \left(2^{101} - 1\right) - \left((1+q)^{101} - 1\right) = 2\alpha^k \left(1 - \left(\frac{1+q}{2}\right)^{101}\right)$$

$$\Rightarrow 2^{101}\left(1 - \left(\frac{1+q}{2}\right)^{101}\right) = 2\alpha^k \left(1 - \left(\frac{1+q}{2}\right)^{101}\right)$$

$\alpha^k = 2^{100}$

$\alpha + k = 102$

74. Let $n(A) = n$, then the

1) Number of symmetric relations is $2^{\left(\frac{n^2+n}{2}\right)}$

2) Number of symmetric and reflexive relations is $2^{\left(\frac{n^2-n}{2}\right)}$

75. $S.D.(x_i) = S.D.(x_i - 10)$

For More Material Join: @JEEAdvanced_2025